Assignment 3

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```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.6
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.1     v tibble    3.3.0
v lubridate 1.9.4     v tidyr     1.3.1
v purrr     1.2.0
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becon
```

```
library(here)
```

```
here() starts at /Users/christinatreacy/Desktop/course stuff/spring 26/ESM 262/262_Assignmen
```

```
library(ggplot2)
```

Part 1: Building the Dataset

```
# Set seed so the random data is reproducible
set.seed(123)

# Create day column
day <- 1:365
```

```r
# Create made-up daily PM2.5 values
# Using rnorm with mean = 15 and sd = 8 gives mostly realistic urban values
# pmax() prevents negative PM2.5 values
pm25 <- pmax(rnorm(365, mean = 15, sd = 8), 0)

# Create empty season column
season <- rep(NA, 365)

# Assign seasons based on day number
season[day >= 1 & day <= 90] <- "winter"
season[day >= 91 & day <= 180] <- "spring"
season[day >= 181 & day <= 270] <- "summer"
season[day >= 271 & day <= 365] <- "fall"

# Convert season to a factor with correct levels
season <- factor(season,
                 levels = c("winter", "spring", "summer", "fall"))

# Create empty risk column for Part 2
risk <- rep(NA, 365)

# Combine into one data frame
air_quality <- data.frame(day, pm25, season, risk)

# View the first few rows
head(air_quality)
```

```
  day     pm25 season risk
1   1 10.51619 winter   NA
2   2 13.15858 winter   NA
3   3 27.46967 winter   NA
4   4 15.56407 winter   NA
5   5 16.03430 winter   NA
6   6 28.72052 winter   NA
```

```r
# Save the data frame as a CSV
write.csv(air_quality, "air_quality.csv")
```

Part 2: Classify Risk with a For Loop

I used rnorm() to simulate daily PM2.5 concentrations with a mean of 15 µg/m³ and standard deviation of 8 µg/m³. This creates mostly low-to-moderate daily values with occasional higher pollution days, which is reasonable for an urban monitoring station. Negative values were replaced with 0 using pmax() because PM2.5 concentrations cannot be negative.

```r
# Create variables to track unhealthy streaks
current_unhealthy_streak <- 0
max_unhealthy_streak <- 0

# Use a for loop to classify each day
for (i in 1:nrow(air_quality)) {

  # Classify PM2.5 risk level
  if (air_quality$pm25[i] < 12) {
    air_quality$risk[i] <- "good"

  } else if (air_quality$pm25[i] <= 35.4) {
    air_quality$risk[i] <- "moderate"

  } else {
    air_quality$risk[i] <- "unhealthy"
  }

  # Track consecutive unhealthy days
  if (air_quality$risk[i] == "unhealthy") {
    current_unhealthy_streak <- current_unhealthy_streak + 1

    if (current_unhealthy_streak > max_unhealthy_streak) {
      max_unhealthy_streak <- current_unhealthy_streak
    }

  } else {
    current_unhealthy_streak <- 0
  }
}
```

Part 3: Seasonal Summary with a For Loop

This loop goes through each season and counts how many days were classified as "unhealthy." The results are stored in a summary data frame, which makes it easier to compare seasonal differences in high PM2.5 risk.

```r
# Create a vector with the season names
season_names <- c("winter", "spring", "summer", "fall")

# Create an empty vector to store the number of unhealthy days per season
unhealthy_counts <- rep(0, length(season_names))

# Name each position in the vector
names(unhealthy_counts) <- season_names

# Use a for loop to count unhealthy days in each season
for (i in 1:length(season_names)) {

  current_season <- season_names[i]

  unhealthy_counts[i] <- sum(
    air_quality$season == current_season &
      air_quality$risk == "unhealthy")
}

# View the result
unhealthy_counts
```

```
winter spring summer   fall
     0      1      0      1
```

```r
# Convert seasonal unhealthy counts into a data frame
seasonal_unhealthy_summary <- data.frame(
  season = names(unhealthy_counts),
  unhealthy_days = as.numeric(unhealthy_counts))

# View summary table
seasonal_unhealthy_summary
```

```
  season unhealthy_days
1 winter              0
2 spring              1
3 summer              0
4   fall              1
```

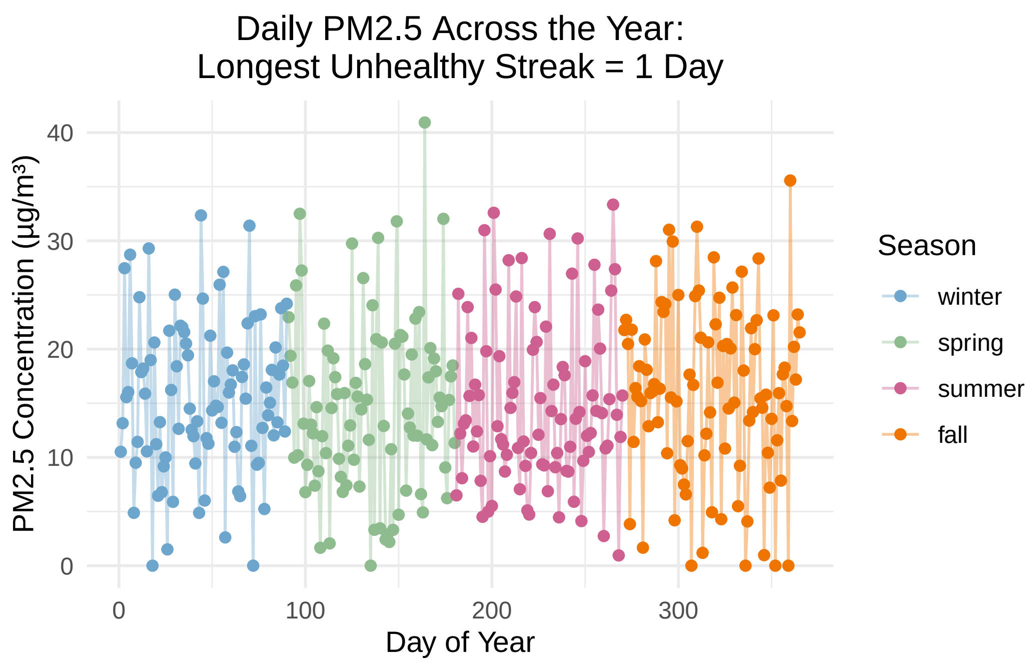
Part 4: Visualize and Report

Plot 1: PM2.5 across the year, colored by season

This shows how PM2.5 changes over the simulated year and makes it easy to see whether certain seasons have higher pollution.

```r
# Create title using sprintf
pm25_title <- sprintf("Daily PM2.5 Across the Year:
Longest Unhealthy Streak = %s Day", max_unhealthy_streak)

# Time series plot of PM2.5
ggplot(air_quality, aes(x = day, y = pm25, color = season)) +
  geom_point() +
  geom_line(alpha = 0.4) +
  labs(title = pm25_title,
    x = "Day of Year",
    y = "PM2.5 Concentration (µg/m³)",
    color = "Season") +
  scale_color_manual(values = c(
      "winter" = "skyblue3",
      "spring" = "darkseagreen",
      "summer" = "hotpink3",
      "fall" = "darkorange2")) +
  theme_minimal()+
  theme(plot.title = element_text(hjust = 0.5))
```



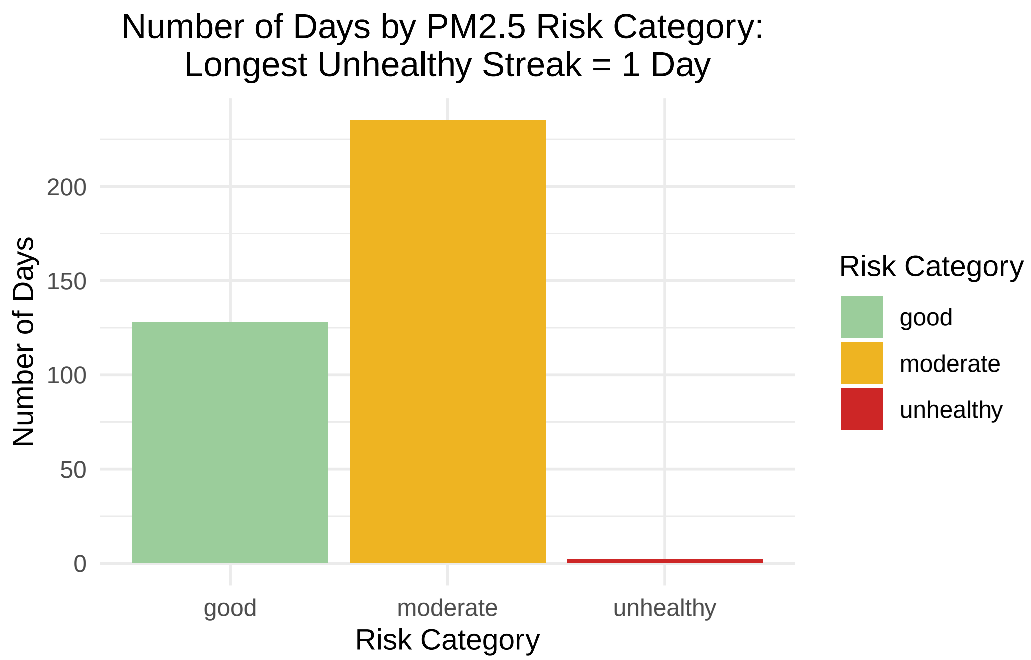
Plot 2: Count of days in each risk category

```
# Create title using sprintf
risk_title <- sprintf("Number of Days by PM2.5 Risk Category:
Longest Unhealthy Streak = %s Day", max_unhealthy_streak)

# Make sure risk categories are ordered
air_quality$risk <- factor(air_quality$risk,
  levels = c("good", "moderate", "unhealthy"))

# Bar chart of risk category counts
ggplot(air_quality, aes(x = risk, fill = risk)) +
  geom_bar() +
  labs(title = risk_title,
    x = "Risk Category",
    y = "Number of Days",
    fill = "Risk Category")+
  scale_fill_manual(values = c(
      "good" = "darkseagreen3",
      "moderate" = "goldenrod2",
      "unhealthy" = "firebrick3")) +
```

```r
  theme_minimal()+
  theme(plot.title = element_text(hjust = 0.5))
```



The simulated air quality data show that most days fell into the good or moderate PM2.5 risk categories, with a smaller number of unhealthy days. The longest consecutive unhealthy streak was only 1 day, suggesting that high-pollution episodes were relatively limited but could still create short-term public health concerns. Seasonal patterns also show that unhealthy days were not evenly distributed across the year, which may indicate that certain seasonal conditions are more favorable for PM2.5 buildup. From a management perspective, even brief unhealthy streaks are important because they can increase exposure risks for sensitive groups, including children, older adults, and people with respiratory conditions. Public health agencies could use this type of monitoring summary to target alerts, reduce outdoor exposure during high-risk days, and evaluate whether seasonal pollution-control strategies are needed.